BSTA551
Statistical Inference I
Lecture Notes

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1 Introduction

Outline:

- Introduction and Definitions
- Notation
- Identifiable Parameters
- Location-Scale Families
- Exponential Families

1.1 Introduction and Definitions

This course is about making inferences from data that are modeled by probability laws. Data arise from an *experiment* that values in a *sample space* Ω.

- Experiment: anything that produces data
- Sample space: set of all possible outcomes of the experiment

Results of experiments are coded into numeric values. *Random variables* (RV) are functions $X : \Omega \to \mathbb{R}$ that map outcomes to numeric values; e.g. a Bernoulli RV can code a binary response.

The *outcome* of an experiment is the collection of values of the random variables defined on the underlying sample space.

The *model* for an experiment is the collection of probability distributions defined on the sample space. Models can be *parametric*, semiparametric, or nonparametric. For example,

$$\mathcal{P} = \left\{ P(X_1 = x_1, \cdots, X_n = x_n; p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} : 0 < p < 1 \right\}$$

is a model for independent and identically distributed Bernoulli random variable from n subjects. Here, the unknown (but fixed) quantity is the probability of response p, and we define the *parameter space* as the set of possible parameter values. In this course, we focus on parametric models where the dimension of parameter space is finite.

Once we have a model, there are two items we need to address

1. Are data consistent with proposed model $\mathcal{P}$?
2. *Assume that the data are generated by a member of $\mathcal{P}$, what can be inferred about the value of the unknown parameters?* [This course.]

For example, consider the simple linear model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,\ \epsilon_i \sim N(0, \sigma^2),$$

where $\epsilon_1, \cdots, \epsilon_n$ are independent.

To address item 1, we need to check for

- linearity of dependence of $\mathrm{E}(Y_i)$ on X_i
- constant variance of ϵ_i
- normality of ϵ_i

To address item 2, we need to answer the following questions:

- Given $\mathcal{P}$, within what limits does β likely lie?
- Given $\mathcal{P}$, are the data consistent with a particular theoretical value for β?

These two questions in item 2 are the concerns of this course (and BSTA552): how to make valid inferences about parameters in fully parametric models, using optimal or at least acceptable procedures. We will focus on **point estimation** of the parameter(s) in this course, and interval estimation and hypothesis testing will be discussed in BSTA552.

1.2 Notation

- $X, X_i, Y_i, \cdots =$ random variables
- $x, x_i, y_i, \cdots =$ observed values of random variables, e.g. $P(X_i = x_i) =$ probability that X_i takes on the value x_i
- $\mathbf{X} = (X_1, \cdots, X_n)$; $\mathbf{x} = (x_1, \cdots, x_n)$
- iid = independent and identically distributed
- $N(\mu, \sigma^2)$: normal distribution with mean μ and variance σ^2
- $\sim$: distributed as, e.g. $X_i \sim N(0, 1)$
- Lowercase Greek letters (θ, λ, etc) are often used for unknown parameters (not always, e.g. Binomial(p))
- Uppercase Greek letters (Θ, Λ, etc) are often used for parameter spaces, e.g. $\Theta =$ parameter space or set of possible values for Θ
- $f(x; \theta)$: the probability density function (pdf) or probability mass function (pmf) of a random variable at parameter $= \theta$, e.g.

$$
\begin{aligned}
f(x;\theta) &= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(x-\theta)^2}{2}\right\}, \; x \in \mathbb{R}, \; \theta \in \mathbb{R} \\
f(x;\theta) &= \begin{cases} e^{-\theta}\theta^x / x! & x = 0, 1, 2, \cdots \\ 0 & \text{otherwise} \end{cases} \quad \theta > 0
\end{aligned}
$$

- $F(x; \theta)$: cumulative distribution functions (cdf)
- $\mathbb{R}^k$: k-dimensional Euclidean space
- $I_A(x) = I(x \in A) = 1_A(x) = 1(x \in A)$: indicator function of set A, i.e.

$$
I(x \in A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}
$$

1.3 Identifiable Parameters

In this course, we assume all parameters of interest (θ) are *identifiable*, i.e. suppose X has model $\{f(x;\theta) : \theta \in \Theta\}$, for any θ, θ', $\theta \neq \theta'$ implies $F(x;\theta) \neq F(x;\theta')$.

Example 1: $X_1, \cdots, X_n$ iid $N(\mu, 1)$. μ is identifiable because any two different values of μ lead to different distributions.

Example 2: $X_1, \cdots, X_n$ iid $N(\mu_1 + \mu_2, 1)$. μ_1 and μ_2 are not identifiable but the sum $\mu_1 + \mu_2$ is identifiable.

1.4 Location-Scale Families

We are used to dealing with parametric families, e.g.

$$\mathcal{F} = \{N(\mu, \sigma^2); -\infty < \mu < \infty, \sigma^2 > 0\}.$$

For many of these, parameters have meanings, e.g. μ is the mean and σ^2 is the variance. It turns out that so many families have similar properties that we can study them in general:

Definition 1. *Let $g(\cdot)$ be any fully specified pdf with continuous distribution. Then if $f_L(x;\mu) = g(x-\mu)$, the family*

$$\mathcal{F} = \{f_L(x;\mu); -\infty < \mu < \infty\}$$

is called the **location family** *generated from $g(\cdot)$ and μ is called the* **location parameter**.

Example:

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

Theorem 1. *(location family).*
a. If $Z \sim g(\cdot)$, then $X = Z + \mu \sim f_L(x; \mu)$;
b. If $X \sim f_L(x; \mu)$, then $X - \mu \sim g(\cdot)$.

Definition 2. *Let $g(\cdot)$ be any fully specified pdf with continuous distribution. For $\sigma > 0$, if $f_S(x; \sigma) = \sigma^{-1} g(x/\sigma)$, the family*

$$\mathcal{F} = \{f_S(x; \sigma); \sigma > 0\}$$

is called the **scale family** *generated from $g(\cdot)$ and σ is called the* **scale parameter**.

Example:

$$g(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right).$$

Theorem 2. *(scale family).*
a. If $Z \sim g(\cdot)$, then $X = \sigma Z \sim f_S(x; \sigma)$;
b. If $X \sim f_S(x; \sigma)$, then $X/\sigma \sim g(\cdot)$.

Definition 3. *Let $g(\cdot)$ be any fully specified pdf with continuous distribution. For $\sigma > 0$, if $f_{LS}(x; \mu, \sigma) = \sigma^{-1} g((x-\mu)/\sigma)$, the family*

$$\mathcal{F} = \{f_{LS}(x; \sigma); -\infty < \mu < \infty, \sigma > 0\}$$

is called the **location-scale family** *generated from $g(\cdot)$. μ is the location parameter and σ is the scale parameter.*

Theorem 3. *(location-scale family).*
a. If $Z \sim g(\cdot)$, then $X = \mu + \sigma Z \sim f_{LS}(x; \mu, \sigma)$;
b. If $X \sim f_{LS}(x; \mu, \sigma)$, then $(X - \mu)/\sigma \sim g(\cdot)$.

Theorem 4. *(location-scale family). If $Z \sim g(\cdot)$ and $E(Z)$ and $Var(Z)$ exist, then for $X = \mu + \sigma Z \sim f_{LS}(x; \mu, \sigma)$,*

$$E(X) = \sigma E(Z) + \mu \text{ and } Var(X) = \sigma^2 Var(Z).$$

Table 1: Examples (Verify)

$g(x)$	type	family
$N(0,1)$	L	$N(\mu, 1)$
	S	$N(0, \sigma^2)$
	LS	$N(\mu, \sigma^2)$
$Exp(1)$	L	$\exp(-(x-\mu))I(x > \mu)$
$g(x) = \exp(-x)I(x > 0)$	S	$\lambda \exp(-\lambda x)I(x > 0)$
	LS	$\lambda \exp(-\lambda(x-\mu))I(x > \mu)$
$U(0,1)$	L	$U(\mu, \mu+1)$
	S	$U(0, \sigma)$
	LS	$U(\mu, \mu+\sigma)$

1.5 Exponential Families

An important type of family of distributions (continuous or discrete) are *exponential families*. The family includes many common distributions and important theorems. First, before we formally define the exponential families, let's define one more term: "support".

Definition 4. *If X is a discrete RV with pmf $f(x)$, the support of X is*

$$supp(X) = \{x : f(x) > 0\}.$$

The support of continuous RV is more difficult to define because continuous densities are not unique. For example, exponential could be defined as $f(x; \lambda) = \lambda e^{-\lambda x} I(x > 0)$ or $f(x; \lambda) = \lambda e^{-\lambda x} I(x \geq 0)$. Both give the same cdf, which is also the same cdf as

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x > 0, x \neq 10 \\ 200 & x = 10. \end{cases}$$

Thus, the generalized definition of "support" that can be applied to continuous RV is:

Definition 5. *Support of the distribution of a random variable X is:*

$$supp(X) = \{x : P(x - h < X < x + h) > 0 \textit{ for all } h > 0\}.$$

That is, x is in the support if the probability of any interval containing x is positive.

Now, let's formally define the exponential families:

Definition 6. *Suppose $f(x;\theta), \theta \in \Theta$ is the pdf (pmf) of the RV X. If f can be expressed in the form*

$$f(x;\theta) = h(x)c(\theta)\exp\left(\sum_{i=1}^{k} w_i(\theta)t_i(x)\right), \tag{1}$$

*where $h(x) \geq 0$, $c(\theta) \geq 0$, $k =$ positive integer, $\{w_1(\cdot), \cdots, w_k(\cdot)\}$ are real-valued linearly independent functions of the parameter(s) θ, $\{t_1(\cdot), \cdots, t_k(\cdot)\}$ are real-valued linearly independent functions of x, and the support is the same for all $\theta \in \Theta$, then $f(x;\theta)$ is called a k-**parameter exponential family** of distributions.*

Examples: Normal, Gamma, Exponential, Beta, Binomial, Poisson.
Counterexamples: Uniform, Laplace.

Note that 1: the representation (1) is not unique; and 2: k is not necessarily equal to dimension of θ

Example: Binomial distribution. Binomial(n, p), $0 < p < 1$.

Example: Poisson distribution. Poisson(λ), $\lambda > 0$.

<u>Example: Normal distribution.</u> $\mathrm{N}(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma^2 > 0$.

Example: not of exponential family form.

1. Uniform distribution;
2.

$$f(x;\theta) = \frac{\exp(x-\theta)}{[1+\exp(x-\theta)]^2}.$$

Now, let's see another example. Let $\mathcal{F}_1 = \{f(x;\theta)|\theta \in \mathbb{R}\}$ where

$$f(x;\theta) = \frac{2\exp\{-(x-\theta)^4\}}{\Gamma(1/4)} \text{ for } x \in \mathbb{R},$$

where $\Gamma(u) = \int_0^\infty t^{u-1}e^{-t}dt$. $\mathcal{F}_1$ is a exponential family, with

- $h(x) = \frac{2\exp(-x^4)}{\Gamma(1/4)}$
- $c(\theta) = \exp(-\theta^4)$
- $w_1(\theta) = 4\theta$, $t_1(x) = x^3$, $w_2(\theta) = -6\theta^2$, $t_2(x) = x^2$, $w_3(\theta) = 4\theta^3$, $t_3(x) = x$

Thus,

$$f(x;\theta) = \frac{2\exp(-x^4)}{\Gamma(1/4)}\exp(-\theta^4)\exp\{4\theta x^3 - 6\theta^2x^2 + 4\theta^3x\},$$

and $\mathcal{F}_1$ is a 3-parameter exponential family, though θ is of dimension 1. Such an exponential family is called a *curved exponential family*.

Definition 7. *A curved exponential family is a family of densities of the form (1) for which the dimension of the vector θ is equal to d and $d < k$. If $d = k$, the family is a full exponential family.*

Now let's redefining parameters:

$$\begin{aligned} \eta_1 &= w_1(\theta) = 4\theta \\ \eta_2 &= w_2(\theta) = -6\theta^2 \\ \eta_3 &= w_3(\theta) = 4\theta^3 \end{aligned}$$

$$\implies f(x;\eta_1,\eta_2,\eta_3) = f(x;\eta) = h(x)c(\eta)\exp(\eta_1x^3 + \eta_2x^2 + \eta_3x), \tag{2}$$

where $c(\eta) = \exp\{-(\eta_1/4)^4\}$. The above parameterization is called *natural parameterization*. Thus, the parameter space Θ was mapped into

$$\mathcal{H}_1 = \left\{\eta = (\eta_1,\eta_2,\eta_3) \mid \begin{array}{ll} \eta_1 = 4\theta, & \eta_2 = 6\theta^2 \\ \eta_3 = 4\theta^3, & \theta \in \mathbb{R}. \end{array}\right\}$$

Note that $\mathcal{H}_1$ is a curve in $\mathbb{R}^3$. Now, we generalize $\mathcal{F}_1$ to a larger family $\mathcal{F}_2 = \{f(x;\eta)|\eta \in \mathcal{H}_2 \subset \mathbb{R}^3\}$, where $f(x;\eta)$ is of the exponential family form (2), but instead of $c(\eta)$, because densities have to integrate to 1 and should equal to

$$c(\eta) = \left(\int h(x) \exp\left\{ \sum_{j=1}^{k} \eta_j t_j(x) \right\} dx \right)^{-1}.$$

The parameter space $\mathcal{H}_2 = \{\eta : c(\eta) > 0\}$ is called the *natural parameter space*. In conclusion, what we've done is

1. $X \in f(x;\theta), \theta = \Theta$ = exponential family
2. Nature parameterization $\eta \in \mathcal{H}_1$
3. Expand $\mathcal{H}_1 \to \mathcal{H}_2$

<u>Example: Poisson distribution.</u> Poisson(λ), $\lambda > 0$.

Exponential families share some desired properties.

Theorem 5. *If X is a RV with pdf or pmf of the form (1) (exponential family), then*

$$
\begin{aligned}
E\left(\sum_{i=1}^{k} \frac{\partial w_i(\theta)}{\partial \theta_i} t_i(X)\right) &= -\frac{\partial}{\partial \theta_i} \log c(\theta) \\
Var\left(\sum_{i=1}^{k} \frac{\partial w_i(\theta)}{\partial \theta_i} t_i(X)\right) &= -\frac{\partial^2}{\partial \theta_i^2} \log c(\theta) - E\left(\sum_{i=1}^{k} \frac{\partial^2 w_i(\theta)}{\partial \theta_i^2} t_i(X)\right).
\end{aligned}
$$

Example: Poisson distribution. Poisson(λ), $\lambda > 0$.

Theorem 6. *For a random sample $X_1, \cdots, X_n$ from an exponential family, the joint distribution also follows an exponential family.*

Theorem 7. *The joint distribution of $t_1(X), \cdots, t_k(X)$ is of exponential family form with the same natural parameters, $w_1(\theta), \cdots, w_k(\theta)$.*

Corollary 1. *For a random sample $X_1, \cdots, X_n$ from an exponential family, the statistics $T_i(X) = \sum_{j=1}^{n} t_i(X_j)$, for $i = 1, \cdots, k$ also follows a k-parameter exponential family.*

Theorem 8. *For any $\mathcal{S} \subset \{1, 2, \cdots, k\}$, the joint distribution of $\{t_i(X), i \in \mathcal{S}\}$, conditional on $\{t_i(X), i \notin \mathcal{S}\}$ is of exponential family form, with a distribution depending only on $\{w_i(\theta), i \in \mathcal{S}\}$.*

Note: Theorem 8 will be important because it will allow inference about selected components of the natural parameter, in the absence of knowledge of the other components.

Theorem 9. *Consider the nature parameters:*

$$f(x;\eta) = h(x)c(\eta)\exp\left(\sum_{i=1}^{k}\eta_i t_i(x)\right).$$

For any η on the interior of the natural parameter space, the moment generating function (mgf) of $(t_1(X), \cdots, t_k(X))$ exists and is given by

$$M_{(t_1(X),\cdots,t_k(X))}(s) = \frac{c(\eta)}{c(\eta+s)}.$$

2 Principle of Data Reduction I

Outline:

- Sufficiency
- Factorization Theorem
- Minimal Sufficient

2.1 Sufficiency

From an experiment, the data we obtained is just a long list of numbers. It contains a lot of information, but the data itself does not have any interpretability. Thus, we want to summarize information in sample by computing *statistics*, which are functions of the sample. We seek method to choose the "right" statistics that

1. Do not discard important about unknown parameter θ.
2. Do discard information that is irrelevant regarding knowledge about θ.

Examples of statistics are:

1. Sample mean: $\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$
2. Order statistics: $(X_{(1)}, X_{(2)}, \cdots, X_{(n)})^T$, where $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$
3. Sample median: $X_{(n+1)/2}$ or $\frac{X_{(n/2)}+X_{(n/2)+1}}{2}$
4. Sample variance: $s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X}^2)}{n-1}$
5. The sample: $\mathbf{X}$
6. The first value sampled: X_1

Let $T(\mathbf{X})$ denote a function of the random variables, $X_1, \cdots, X_n$, with observed value $T(\mathbf{x})$. If we observe a different sample $y_1, \cdots, y_n$, we say the statistic $T(\mathbf{X})$ is a form of *data reduction* if $T(\mathbf{x}) = T(\mathbf{y})$.

In general, we follow three principles as the basis for inference:

1. *Sufficiency:* method that does not discard information about θ, while reducing the data.
2. *Likelihood:* function of parameter from observed sample that contains all information about θ available from the sample.
3. *Invariance:* inference procedure should be measurement invariant (as long as model for data is preserved).

We first talk about sufficiency. The *sufficiency principle* states that if $T(\mathbf{X})$ is a *sufficient statistic* for the parameter of interest θ, then any information about θ should depend on $\mathbf{X}$ through $T(\mathbf{X})$. For example, $\mathbf{X} = (X_1, \cdots, X_n)$, $X_i \sim N(\theta, 1)$: later we will show that $\bar{X}$ is a sufficient statistic of θ, and we should focus on $\bar{X}$ (1 dimension) instead of $\mathbf{X}$ (n dimensions).

We formally define the sufficient statistic in the following.

Definition 8. *Let $f(\mathbf{x}; \theta)$ be the joint pdf (pmf) of the sample $\mathbf{X}$. We say a statistic $T(\mathbf{X})$ is a sufficient statistic for θ if $f(\mathbf{x}|T = T(x); \theta) = g(\mathbf{x})$.*

We can understand the sufficient statistic as follows: suppose an investigator reports the value of $T(\mathbf{X})$, but discards the original dataset. If T is sufficient, we can construct an equivalent sample $\mathbf{X}'$ through the conditional distribution $g(\mathbf{x})$ (using a random number generator).

Example: Poisson distribution. Let $X_1, \cdots, X_n$ iid Poisson(λ). Show that $T(\mathbf{X}) = \sum_{i=1}^{n} X_i$ is a sufficient statistic of λ.

Example: Normal distribution. Let X_1, X_2 iid N(θ, 1). Show that $T(\mathbf{X}) = \bar{X}$ is a sufficient statistic of θ. (Hint: $(X_1, X_2)|\bar{X}$ does not have a pdf on $\mathbb{R}^2$)

2.2 Factorization Theorem

From the above examples, we can see that, in general, checking sufficiency directly is difficult since we need to 1. compute the conditional distribution of $\mathbf{X}|T(\mathbf{X})$, and 2. "guess" a candidate statistic. Also, because the definition of sufficient statistic is not prescriptive, sufficient statistic is never unique (original data are always sufficient but not helpful).

In this section, we study the *Factorization Theorem*, which provides a convenient way to identify sufficient statistics directly from the pdf/pmf.

Theorem 10. *Let $\mathbf{X}$ be a random vector with pdf/pmf $f(\mathbf{X};\theta), \theta \in \Theta$. A statistic $T(\mathbf{X})$ is sufficient for θ if and only if there exist $g(T(\mathbf{x});\theta)$ and $h(\mathbf{x})$ such that*

$$f(\mathbf{X};\theta) = g(T(\mathbf{x});\theta)h(\mathbf{x}) \tag{3}$$

for all $\mathbf{x}$ and θ.

Discrete case:

We will not show the proof of continuous case because it involves higher level mathematics about conditional distribution.

Example: Poisson distribution.

Example: Normal distribution.

Example: Exponential families.

$$f(x_i;\theta) = h(x)c(\theta)\exp\left\{\sum_{j=1}^{k} w_j(\theta)t_j(x)\right\},\ i = 1, \cdots, n.$$

For iid samples from smooth densities with support independent of θ, subject to some regularity conditions, exponential families are the only distributions that have k-dimensional sufficient statistics for all sample sizes. The following result is the formal theorem for the 1-parameter case.

Theorem 11. *Let $X_1, \cdots, X_n$ be iid, with pdf (pmf) $f(x;\theta)$ in the model $\mathcal{P} = \{f(x;\theta) : \theta \in \Theta \subset \mathbb{R}\}$. Suppose that the support of the distributions in $\mathcal{P}$ is independent of θ, that Θ contains an open interval, and that the pdfs in $\mathcal{P}$ are all differentiable as functions of x for each θ. If for all n there is a 1-dimensional sufficient statistic for θ, then $\mathcal{P}$ must be a 1-parameter exponential family.*

Notes: *Non-uniqueness* of sufficient statistic is apparent from Factorization Theorem. For example, suppose $X_i \sim \ iid\ N(0, \sigma^2)$. Then,

$$f(\mathbf{x}; \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp(-\sum_{i=1}^{n} x_i^2/\sigma^2).$$

By Factorization Theorem, $T(X) = \sum_{i=1}^n X_i^2, \exp(\sum_{i=1}^n x_i^2), (\sum_{i=1}^n x_i^2)^2$, etc., are all sufficient. In fact, any 1-to-1 function of $\sum_{i=1}^n x_i^2$ is sufficient.

2.3 Minimal Sufficiency

In any problem there are many sufficient statistic. E.g., if $\sum_i X_i$ is a sufficient statistic for θ, then so is $(\sum_i X_i, \sum_i X_i^2)$. If no further reduction from the statistic $T(\mathbf{x})$ is possible (while retaining sufficiency), then $T(\mathbf{x})$ is **minimal sufficient**. Of all sufficient statistics, a minimal sufficient statistic provides the greatest possible reduction of the data.

Definition 9. *If $T(\mathbf{x})$ is sufficient for θ, then it is* **minimal sufficient statistic (MSS)** *if for any other sufficient statistic $T^*(\mathbf{x})$, $T(\mathbf{x})$ is a function of $T^*(\mathbf{x})$, i.e. $T(\mathbf{x}) = g(T^*(\mathbf{x}))$ for some function g.*

Remark 1. *By requiring that $T(\mathbf{x}) = g(T^*(\mathbf{x}))$, we are making it* **data reduction** *from $T^*(\mathbf{x})$.*

Remark 2. *MSS is never unique: any 1-to-1 transformation of MSS is also MSS.*

Remark 3. *Conceptually MSS is usefull, but it doesn't tell us:*

1. *How to verify if a statistic is a MSS.*
2. *If a MSS exists.*
3. *How to find a MSS.*

Fortunately, the following theorem proposed by Lehmann and Scheffé (Casella and Berger Theorem 6.2.13) can help us identify a MSS.

Theorem 12. *Let $X \sim f(x;\theta)$. Suppose there is a $T = T(\mathbf{x})$ such that for two sample $\mathbf{x}$ and $\mathbf{y}$ in the sample space, the ratio*

$$g(\mathbf{x}, \mathbf{y}; \theta) = \frac{f(\mathbf{x};\theta)}{f(\mathbf{y};\theta)} \text{ is independent on } \theta \text{ iff } T(\mathbf{x}) = T(\mathbf{y}). \tag{4}$$

Then T is a MSS.

Proof:

(cont.)

Operationally, the theorem can verify if a candidate is MSS by verifying (4); it can also help find one, by calculating the ratio $g(\mathbf{x}, \mathbf{y}; \theta)$ and manipulating it to try to find T such that the ratio does not depend on θ iff $T(\mathbf{x}) = T(\mathbf{y})$.

Example: $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid $Poisson(\lambda)$.

Example: $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid $N(\mu, \sigma^2)$.

Theorem 13. *(exponential families) If $X_1, \cdots, X_n$ are iid with*

$$f(x_i;\theta) = h(x)c(\theta)\exp\left\{\sum_{j=1}^{k} w_j(\theta)t_j(x)\right\},$$

then $\mathbf{T} = (T_1, \cdots, T_k)$ is a MSS, where $T_j = T_j(\mathbf{x}) = \sum_{i=1}^{n} t_j(x_i)$.

3 Principle of Data Reduction II

Outline:

- Ancillary Statistics
- Complete Families
- Likelihood Principle

3.1 Ancillary Statistics

Ancillarity is the dual of sufficiency; sufficiency describes where all the information in the data is contained, whereas ancillarity describes where there is no information. Ancillary statistics may not exist, they may exist but be hard to find, and they may not be unique.

Definition 10. $\mathbf{X} = (X_1, \cdots, X_n)$, *$X_i$ iid with pdf (pmf) $f(x;\theta)$. A statistic, $S(\mathbf{x})$, whose distribution does not depend upon the model parameter, θ, is called an* **ancillary statistic**.

Example: (location family) $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid from a location family: $X_i \sim f_L(x;\theta) = g(x-\theta)$. Then $R = X_{(n)} - X_{(1)}$ is ancillary.

Example: (scale family) $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid from a scale family: $X_i \sim f_S(x;\sigma) = g(x/\sigma)/\sigma$. Let $S_j = X_j/X_n$. Then any function of $\mathbf{S} = (S_1, \cdots, S_n)$ is ancillary for σ.

In general, suppose that there is a statistic $S = s(\mathbf{X})$ and an ancillary statistic $C = c(\mathbf{X})$ so that $T = (S, C)$ taken together are minimal sufficient. Then the **ancillarity principle**, sometimes also called **conditionally principle**, states: inference on the parameter should be based on the conditional distribution of S, given the ancillary statistic C.

Note, if C is ancillary, the joint density of (S, C) is proportional to the conditional density of S given C

$$f_{S,C}(s, c; \theta) = f_{S|C}(s|c; \theta) f_C(c; \theta) \propto f_{S|C}(s|c; \theta).$$

Another way to understand conditionally principle is through *nuisance parameters*: suppose that we have a nuisance parameter λ and a parameter of interest ψ, with $\theta = (\psi, \lambda)$. If $T = (S, C)$ is minimal sufficient for θ but C is ancillary for ψ (but depends on λ), then C is ancillary if λ is held fixed. Using the conditionally principle, we can make inference on ψ from the conditional distribution of S given C.

There are some difficulties with ancillary statistics:

- No general method for constructing C
- In general can be hard to find
- Not unique: conditioning on different statistics might give different conditional inferences
- Conditional distributions might be complicated

3.2 Complete Families

Definition 11. *Let $f(t; \theta)$ be a family of statistic of pdf's/pmf's for a statistic, $T = T(\mathbf{X})$ (not necessarily sufficient statistic), $\theta \in \Theta$. If for any function $g(\cdot)$, we have that $E(g(T)) = 0$ implies $P(g(T) = 0) = 1$, then the family for T is called* **complete**. *In addition, T is called* **complete statistic**.

This definition says: T is complete $\implies$ only function of T that can have 0 expectation for all θ is the zero function. In other word, there *can not* be any non-zero unbiased estimates of 0 based on T.

Example: $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid Bernoulli(p). $T = \sum_{i=1}^{n} X_i$ is complete.

Example: Statistics that are *not complete*:

- $T = \mathbf{X} = (X_1, \cdots, X_n)$ with X_i iid
- $T = \bar{X}$ with X_i iid $N(0, \sigma^2)$, σ^2 unknown

Note that for the second one, $\bar{X}$ is complete for the $N(\theta, 1)$ family.

Remark 4. *If T is sufficient, then it contains all information regarding θ. If T is also complete, it says T contains no irrelevant information about θ.*

Note: A MSS that is not complete may also contain only relevant information.

Theorem 14. *If T is complete and sufficient and a minimal sufficient statistic exists, then T is minimal sufficient.*

Next, let's explore the main result for identifying complete, sufficient statistics.

Theorem 15. *If $X_1, \cdots, X_n$ iid, $X_i \sim$ exponential family with pdf (pmf)*

$$h(x)c(\theta)\exp\left\{\sum_{j=1}^{k} w_j(\theta)t_j(x)\right\},$$

and if $\mathcal{H} \equiv \{(w_1(\theta), \cdots, w_k(\theta))|\theta \in \Theta\}$ contains an open rectangle in $\mathbb{R}^k$, then $T = (T_1, \cdots, T_k)$ is a complete and sufficient statistic, where $T_j = \sum_{i=1}^{n} t_j(x_i)$.

The proof of Theorem 15 is complex and not done here (can be found in Lehmann and Romano, 2005, *Testing Statistical Hypothesis*, Section 4.3).

Theorem 15 can not be applied to curved exponential families ($k > \dim(\theta)$) because $\mathcal{H}$ does not contain an open rectangle. Curved exponential families in general do not have the same relationship among sufficiency, minimal sufficiency and completeness as full rank exponential families. Examples of curved exponential families are $N(\mu, \mu^2)$ (normal approximation of Poisson distribution), or the one illustrated on Page 1-13, Section 1.5.

Another import result is the Basu's Theorem, which allows us to deduce independence between S and T without needing to find their joint distribution.

Theorem 16. *(Basu's Theorem) If T is a complete and sufficient statistic, then T is independent of any ancillary statistic.*

Example: $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim \text{Exp}(\theta)$. Compute the expect value for the statistic

$$S(\mathbf{X}) = \frac{X_n}{X_1 + X_2 + \cdots + X_n}.$$

Example: $\mathbf{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim N(\mu, \sigma^2)$, σ^2 is known. Show $\bar{X}$ and S^2 are independent.

3.3 Likelihood Principle

We will not discuss this part in great detail except to introduce the likelihood function (needed later) and to briefly discuss the likelihood principle.

Definition 12. *Suppose a sample* $\mathbf{X} = (X_1, \cdots, X_n)$ *has joint pdf/pmf* $f(\mathbf{x}; \theta)$, $\theta \in \Theta$. *Given an observed outcome* $\mathbf{X} = \mathbf{x}$, *the* **likelihood function** *is the function* $L : \theta \to \mathbb{R}$ *given by*

$$L(\theta) \equiv L(\theta; \mathbf{x}) = f(\mathbf{x}; \theta).$$

The **log-likelihood function** *is* $l(\theta) = \log L(\theta)$.

Comments:

1. The likelihood function is a function of θ with $\mathbf{x}$ fixed
2. Given $\mathbf{x}$, if $L(\theta_2; \mathbf{x}) > L(\theta_1; \mathbf{x})$, then the data are more likely to have occured under θ_2 than under θ_1. In other word, θ_2 is considered more consistent with the data than θ_1
3. Usually interested in relative magnitudes, so factors involving only $\mathbf{x}$ can be dropped
4. If T is sufficient, then by the Factorization theorem, $f(\mathbf{x}; \theta) = g(T(\mathbf{x}); \theta)h(\mathbf{x})$, so $L(\theta; \mathbf{x}) \propto g(T(\mathbf{x}); \theta)$. (The likelihood depends on the data only through a sufficient statistic)
5. The likelihood function is usually used to generate estimators (e.g., maximum likelihood estimators)

Let $\mathbf{x}$ and $\mathbf{y}$ are two sample points. From the Factorization theorem, if T is sufficient with $T(\mathbf{x}) = T(\mathbf{y})$, then

$$L(\theta; \mathbf{x}) = g(T(\mathbf{x}); \theta)h(\mathbf{x}) = g(T(\mathbf{x}); \theta)h(\mathbf{y})\frac{h(\mathbf{x})}{h(\mathbf{y})} \propto L(\theta; \mathbf{y}).$$

The **weak likelihood principle** states that if

$$L(\theta; \mathbf{x}) \propto L(\theta; \mathbf{y}) \text{ as function of } \theta,$$

then the inference about θ should be the same for $\mathbf{x}$ and $\mathbf{y}$.

Note: because of the close connection between $T(\mathbf{x}) = T(\mathbf{y})$ and $L(\theta; \mathbf{x}) \propto L(\theta; \mathbf{y})$, within a given experiment, the sufficiency and weak likelihood principles are essentially equivalent.

Now, let's discuss the **formal likelihood principle**. Suppose now there are two different experiments that could be used to draw inferences on θ. The first produces data $\mathbf{X}_1$ with pdf/pmf $f_1(\mathbf{x}_1; \theta)$, and the second produces $\mathbf{X}_2$ with pdf/pmf $f_2(\mathbf{x}_2; \theta)$. The parameter space Θ is the same for both experiments. The formal likelihood principle states that if $\mathbf{x}_1$ is an outcome from the first experiment and $\mathbf{x}_2$ is an outcome from the second such that

$$L(\theta; \mathbf{x}_1) \propto L(\theta; \mathbf{x}_2),$$

then the inferences about θ should be the same. That is, inference should depend on the data only through the likelihood function.

This means that the same conclusions would be made from samples drawn from different probability spaces, i.e., with different sampling designs, if the resulting likelihood functions are proportional. The conditionality principle in this setting means that if we randomly select which experiment to do, that we should condition on the experiment actually done, and not consider that a different experiment might have been done. Birnbaum's theorem (CB Theorem 6.3.6) states that the formal likelihood principle is equivalent to the combination of the sufficiency principle and the conditionality principle.

Example: Consider a sequence of iid Bernoulli trials $X_1, X_2, \cdots$, for the purpose of investigating $p = P(X_i = 1)$. In experiment 1, we fix n, the total number of trials. Here, $T = \sum_{i=1}^{n} X_i$ is a complete and sufficient statistic (CSS), with $T \sim \text{Bin}(n, p)$. In experiment 2, we fix s, the total number of success, and continue sampling until $\sum_i X_i = s$. Here, the total number of trials, N, is random (negative Binomial distribution), with

$$P(N = n) = \binom{n-1}{s-1} p^s (1-p)^{n-s}$$

If we observe $T = s$ in the first experiment and $N = n$ in the second, then both experiments give s successes out of n trials, and the likelihood in both cases is $\propto p^s(1-p)^{n-s}$, If we follow the *formal likelihood principle*, then the inference should be the same.

However, frequentist inference often does not follow the likelihood principle. In our example, for experiment 1, T/n is an unbiased estimator of p; but for experiment 2, s/N is a biased estimator of p. Moreover, inference usually requires more than a point estimator (variance of estimator depend on the sampling framework). Thus, even when inference is based on the distribution of the sufficient statistic, different assumptions about the sampling distribution can lead to different inferences.

4 Methods of Estimation

Outline:

- Method of Moments
- Maximum Likelihood
  - Multiple Parameters
  - Invariance Property
  - Exponential Families
- Other Methods

4.1 Introduction

In this section, we will mostly assume $\boldsymbol{X} = (X_1, \cdots, X_n)$ with X_i iid, $X_i \sim f(x_i; \theta)$, $\theta \in \Theta$. As we discussed in previous sections, Θ determines the behavior of $\boldsymbol{X}$. From observed $\boldsymbol{X}$, we want to estimate θ, usually denoted as $\hat{\theta}$ or $\tilde{\theta}$.

A **point estimator** is any function $T(\boldsymbol{X})$ of the data that targets a population quantity, $\tau(\theta)$. An **estimate** is the realized value, $T(\boldsymbol{x})$. Based on this definition, there's no mention of closeness of the estimator to the target, but of course, this desirable.

Some general methods of finding estimators:

1. Method of Moments (MOM)
2. Maximum Likelihood Estimator (MLE)
3. Bayes methods
4. Unbiased Estimating Equations

4.2 Method of Moments (MOM)

The method is popularized by Karl Pearson in late 1800's. The basic idea of MOM is to equate population quantities to empirical quantities.

Define

$$\mu_r \equiv \mu_r(\theta) = \mathrm{E}(X_i^r),\ r = 1, 2, \cdots$$

Since $\mu_1, \mu_2, \cdots$ are functions of θ, and we can estimate $\mathrm{E}(X_i^r)$ by

$$m_r = \frac{1}{n}\sum_{i=1}^{n} X_i^r \text{ (sample } r^{th} \text{ moment; } m_1 = \bar{X},\ m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2),$$

m_r is an estimate of $\mu_r(\theta)$ and we could determine value of $\hat{\theta}$ by solving the equation(s) $\mu_r(\theta) = m_r$.

Definition 13. *Method of Moments (MOM). Suppose $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_k)$. A MOM estimator of θ is given by the solution (in θ) of the k equations*

$$\begin{aligned} m_1 &= \mu_1(\theta) \\ m_2 &= \mu_2(\theta) \\ &\vdots \\ m_k &= \mu_k(\theta). \end{aligned}$$

The method can generalize to use functions of data, i.e. $\sum_i g(X_i)/n = E_\theta(g(X_1))$; or central moments, e.g. $\sum_{i=1}^n (X_i - \bar{X})^2/(n-1) = \mathrm{Var}_\theta(X_i)$.

Example: Exponential MOM. $X_1, \cdots, X_n$ iid $\text{Exp}(\lambda)$.

Example: Normal MOM. $X_1, \cdots, X_n$ iid $N(\mu, \sigma^2)$.

Example: Poisson MOM. $X_1, \cdots, X_n$ iid Poisson(λ).

Example: $X_1, \cdots, X_n$ iid $f(x;\theta) = \theta x^{-2}, 0 < \theta \leq x < \infty$.

MOM is intuitively reasonable, yet can lead to $\hat{\theta} \notin \Theta$.

Example: Binomial MOM. $X_1, \cdots, X_n$ iid Binomial(N, p), both N and p unknown.

Notes: The MOM is a simple procedure to give us an approach for finding an estimator. However,

1. There may not be a solution to MOM equations.
2. MOM estimators are not invariant to transformation of the data or to transformation of θ.
3. May be useful in providing simple estimates of parameters from complex distributions. Also may offer a good starting point for more sophisticated (e.g., iterative) methods.
4. Sufficiency? Likelihood principle?

4.3 Maximum Likelihood

Assume again that $\boldsymbol{X} = (X_1, \cdots, X_n)$ with X_i iid, $X_i \sim f(x_i; \theta)$. The pdf/pmf of $\boldsymbol{X}$ is

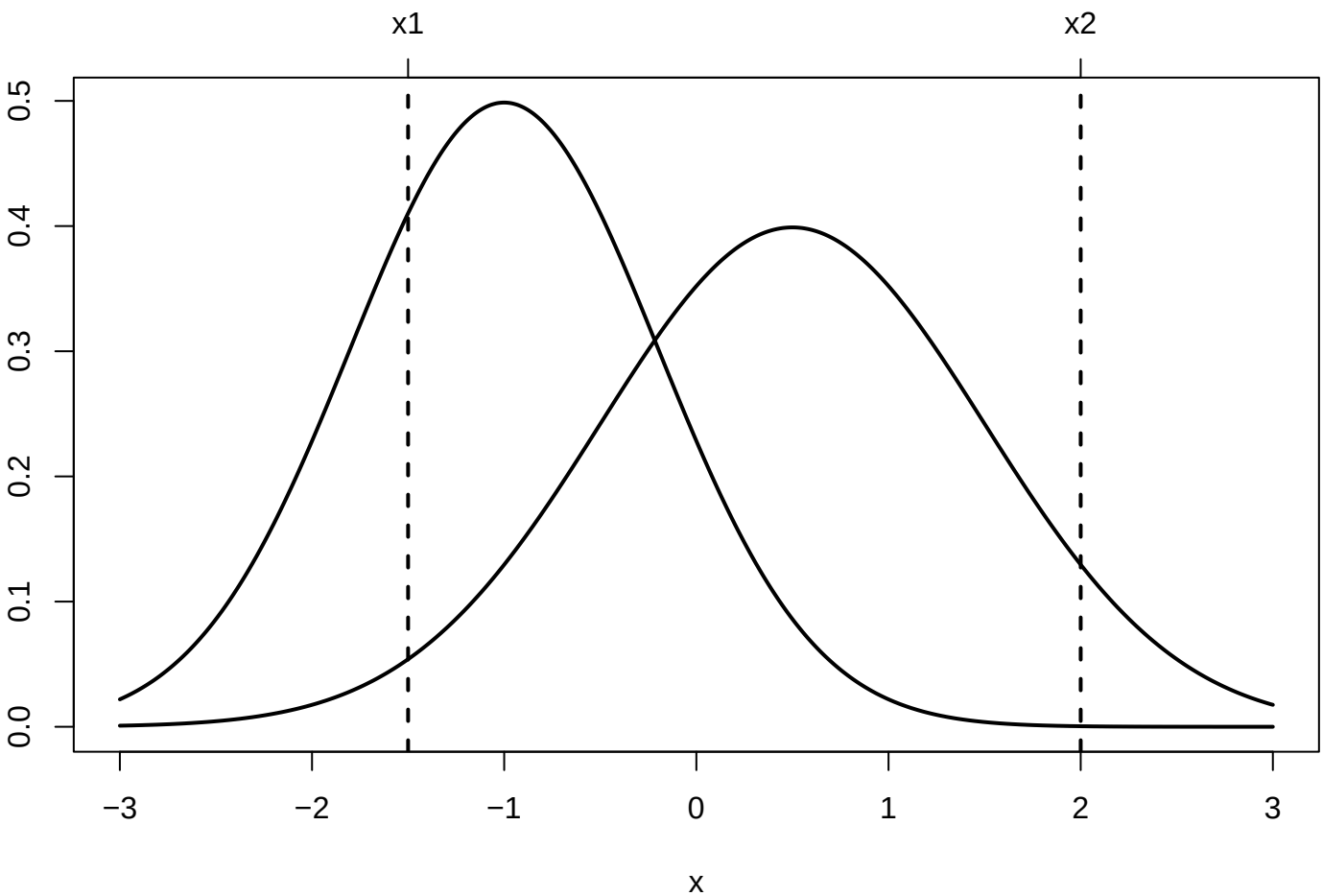
$$f(\boldsymbol{x}; \theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

As a pdf/pmf, the above function is viewed as a function of $(x_1, \cdots, x_n)$, for a given θ. As a likelihood function,

$$L(\theta; \boldsymbol{x}) = f(\boldsymbol{x}; \theta) = \prod_{i=1}^{n} f(x_i; \theta) \tag{5}$$

is a function of θ for an observed set of data points $\boldsymbol{x}$.

Figure 1: Idea behind maximum likelihood estimation: Suppose that $\Theta = \{\theta_1, \theta_2\}$, and that $f(x;\theta_1)$ and $f(x;\theta_2)$ are as follows:



If we observe $X = x_2$, we think it is more likely that X from $f(x;\theta_2)$: it could come from $f(x;\theta_1)$, but for that it would be an outlier. So probably, if $X = x_2$, we would want to estimate $\hat{\theta} = \theta_2$.

Now consider estimating θ.

Definition 14. *For any value of $\boldsymbol{X} = \boldsymbol{x}$, let $\hat{\theta}(\boldsymbol{x})$ be value of θ which maximizes $L(\theta;\boldsymbol{x})$ (equation (5)). The $\hat{\theta} \equiv \hat{\theta}(\boldsymbol{X})$ is the* **Maximum Likelihood Estimator (MLE)** *of θ. That is,*

$$L(\hat{\theta}(\boldsymbol{X});\boldsymbol{X}) = \max\{L(\theta;\boldsymbol{X}) : \theta \in \Theta\}.$$

Note:

- $\hat{\theta}$ is a statistic
- $\hat{\theta} \in \Theta$ (not true for MOM)
- $\hat{\theta}$ has good properties (later)
- $\hat{\theta}$ is not necessarily unique
- $\hat{\theta}$ does not always exist
- Optimizing $\hat{\theta}$ can be computational difficult and numerically sensitive

Note that any monotonic transformation of $L(\theta; \boldsymbol{x})$ will be maximized by the same value(s) as $L(\theta; \boldsymbol{x})$. We often elect to maximize the **log likelihood** function

$$l(\theta; \boldsymbol{x}) = \log\{L(\theta; \boldsymbol{x})\} = \sum_{i=1}^{n} \log\{f(x_i; \theta)\} \text{ (last expression if iid).}$$

For most distributions, $l(\theta; \boldsymbol{x})$ involves simpler expression than $L(\theta; \boldsymbol{x})$ (e.g. exponential families) and we will focus on $l(\theta; \boldsymbol{x})$.

When our likelihood functions are well behaved (e.g., continuous, differentiable with no restrictions on the parameters' parameter space) we can use calculus tools to find the global maximums. Under these conditions, possible candidates for the MLE include solutions of the *score functions:*

$$\frac{\partial l(\theta; \boldsymbol{x})}{\partial \theta} = \mathbf{0}$$

There are additional checks that may be needed to insure these solutions yield a global maximum. We can often check second derivatives (being negative). Sometimes we need to check boundary values as well. Note that solutions to the score equations could yield local minimums or maximums, global minimums or maximums, or inflection points.

Example: Exponential MLE. $X_1, \cdots, X_n$ iid $\text{Exp}(\lambda)$.

Example: Bernoulli MLE. $X_1, \cdots, X_n$ iid Bernoulli(p), $0 < p < 1$.

Example: Poisson MLE. $X_1, \cdots, X_n$ iid Poisson(λ).

4.3.1 Multiple Parameters

When $\theta = (\theta_1, \cdots, \theta_k)^T$ is a vector, we still define $L(\theta; \boldsymbol{x}) = f(\boldsymbol{x}; \theta)$ and $l(\theta; \boldsymbol{x}) = \log\{L(\theta; \boldsymbol{x})\}$. The vector of partial derivatives (or gradient) and matrix of second partial derivatives (Hessian matrix, $\mathbf{H}$) can be denoted

$$\nabla l(\theta; \boldsymbol{x}) = \frac{\partial}{\partial \theta} l(\theta; \boldsymbol{x}) = \begin{pmatrix} \frac{\partial l(\theta;\boldsymbol{x})}{\partial \theta_1} \\ \vdots \\ \frac{\partial l(\theta;\boldsymbol{x})}{\partial \theta_k} \end{pmatrix}$$

and

$$\nabla^2 l(\theta; \boldsymbol{x}) = \frac{\partial}{\partial \theta \partial \theta^T} l(\theta; \boldsymbol{x}) = \begin{pmatrix} \frac{\partial^2 l(\theta;\boldsymbol{x})}{\partial \theta_1^2} & \cdots & \frac{\partial^2 l(\theta;\boldsymbol{x})}{\partial \theta_1 \partial \theta_k} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 l(\theta;\boldsymbol{x})}{\partial \theta_k \partial \theta_1} & \cdots & \frac{\partial^2 l(\theta;\boldsymbol{x})}{\partial \theta_k^2} \end{pmatrix}$$

Here θ^T denotes the transpose of θ. If the vector nature of the parameters needs to be emphasized, then they may be set in bold; e.g. $\frac{\partial}{\partial \boldsymbol{\theta}} l(\boldsymbol{\theta}; \boldsymbol{x})$.

Notes:

- The vector of maximum likelihood estimates $\hat{\theta}$ is still defined as the value of θ maximizing $L(\theta; \boldsymbol{x})$
- If the likelihood function is smooth, then candidate points for the MLEs can be identified as the solutions to the equations $\nabla l(\theta; \boldsymbol{x}) = 0$ (k equations in k unknowns).
- A solution is a local max if $-\nabla^2 l(\hat{\theta}; \boldsymbol{x})$ is positive definite (pd)
  - A symmetric matrix $\mathbf{H}$ is pd if $\boldsymbol{x}^T \mathbf{H} \boldsymbol{x} > 0$ for all $\boldsymbol{x} = 0$
  - An equivalent condition is that all eigenvalues of $\mathbf{H}$ are positive
  - Another equivalent condition is that the determinants of the leading $j \times j$ submatrices are > 0 for $j = 1, \cdots, k$
- Often MLE are calculated by applying a numerical optimization algorithm (except for some simple problems) directly to $l(\theta; \boldsymbol{x})$ (a numerical test for positive definiteness can be applied to $-\nabla^2 l(\hat{\theta}; \boldsymbol{x})$)

Example: Normal MLE. $X_1, \cdots, X_n$ iid $N(\mu, \sigma^2)$.

Example: logistic regression. $X_1, \cdots, X_n$ independent with Bernoulli(p_i), where

$$p_i = P(X_i = 1) = \frac{\exp(\alpha + \beta z_i)}{1 + \exp(\alpha + \beta z_i)}$$

and z_i are known constants, e.g. $z_i =$ age of subject i.

4.3.2 Invariance Property

Theorem 17. Invariance property of MLE's. *Suppose a distribution is indexed by θ, but an estimator $\tau(\theta)$ is desired. The* **Invariance property** *says that if $\hat{\theta}$ is the MLE for θ, then $\tau(\hat{\theta})$ is the MLE for $\tau(\theta)$. That is, if $\eta = \tau(\theta)$, then $\hat{\eta} = \tau(\hat{\theta})$.*

Proof. Case 1: $\tau(\theta)$ is a 1-1 function of θ.

Proof. Case 2: general $\tau(\theta)$.

From the above proof, we can see that the invariance property is very general and can be applied to a wide class of functions; e.g.

- 1-1 functions: $\tau(\theta) = e^{\theta} \to$ MLE: $e^{\hat{\theta}}$
- Non 1-1 functions: $\tau(\theta) = \theta^2 \to$ MLE: $\hat{\theta}^2$
- More complicated functions: $\tau(\theta) = \sqrt{\theta(1-\theta)} \to$ MLE: $\sqrt{\hat{\theta}(1-\hat{\theta})}$

4.3.3 Exponential Families

For nature parameter space, let $X_1, \cdots, X_n$ iid, $X_i \sim$ exponential family with pdf (pmf)

$$h(x_i)c(\theta)\exp\left\{\sum_{j=1}^{k} w_j(\theta)t_j(x_i)\right\},$$

the likelihood function can be written as

$$L(\theta;\boldsymbol{x}) = h(\boldsymbol{x})c^n(\theta)\exp\left\{\sum_{j=1}^{k} w_j(\theta)T_j(\boldsymbol{x})\right\} = h(\boldsymbol{x})c^*(\eta)\exp\left\{\sum_{j=1}^{k} \eta_j T_j(\boldsymbol{x})\right\},$$

where $T_j(\boldsymbol{x}) = \sum_{i=1}^{n} t_j(x_i)$, and

$$l(\eta;\boldsymbol{x}) = \log h(\boldsymbol{x}) + \log c^*(\eta) + \sum_{j=1}^{k} \eta_j T_j(\boldsymbol{x}),$$

so that the MLE for $\eta_1, \cdots, \eta_k$ is solution to the k equations:

$$T_j(\boldsymbol{x}) = -\frac{\partial}{\partial \eta_j}\log c^*(\eta) = \mathrm{E}_\eta T_j(\boldsymbol{x}).$$

(from Page 1-15, Section 1, Theorem 5). Also from Theorem 5, (simplified for the case of natural parameters),

$$-\nabla^2 l(\eta;\boldsymbol{x}) = -\frac{\partial^2}{\partial\eta\partial\eta^T}\log c^*(\eta) = \mathrm{Var}((T_1(\boldsymbol{x}), \cdots, T_k(\boldsymbol{x}))),$$

which is pd, so is unique and maximizes $l(\eta;\boldsymbol{x})$.

4.4 Other Methods

Note: Bayesian inference will be discuss later in the course.

There exists several other approach to estimation (sometimes equivalent to MOM, MLE, Bayes), e.g.

- For categorical data: minimize chi-square goodness-of-fit statistic
- Least squares
- Estimating equation methods (will not discuss in this course, but very commonly used in Biostatistics)

Example: Regression.

$$Y_i = \alpha + \beta x_i + \epsilon_i.$$

Least square estimation: find α, β which minimize

$$\sum_i [Y_i - (\alpha + \beta x_i)]^2.$$

Example: Estimating equation. For $U(\boldsymbol{X}, \theta)$ such that $\mathrm{E}_\theta(U(\boldsymbol{X}, \theta)) \equiv 0$, find $\tilde{\theta}$ such that $U(\boldsymbol{X}; \tilde{\theta}) = 0$.

5 Evaluating Estimators I: Consistency and MSE

Outline:

- Criteria for Evaluating Estimators
- Consistency
- Mean Squared Error

5.1 Criteria for Evaluating Estimators

Consider the following setting, where we observe the sample

$$\mathbf{X} = (X_1, \ldots, X_n),$$

where $\mathbf{X}$ is a random drawn from the population model parameterized by θ, with

$$\mathbf{X} \sim f(\mathbf{x}|\theta), \qquad \theta \in \Theta.$$

We denote $\hat{\theta}$, an estimator of θ, as

$$\hat{\theta} = \hat{\theta}(\mathbf{X}).$$

Note:

- Estimator is function of the data only, not (also) of θ
- θ, or some function $\tau(\theta)$, is the unknown we are trying to estimate

Now suppose that we have obtained some estimator $\hat{\theta}$. How do we judge how "good" this estimator is? Let $\mathcal{L}(\hat{\theta})$ be the distribution or law of $\hat{\theta}$. To answer this question, we need to look at $\mathcal{L}(\hat{\theta})$ or properties of $\mathcal{L}(\hat{\theta})$.

A good estimator should usually be close to the true value:

1. Finite (small sample) properties: Is $\hat{\theta}$ unbiased? Is $\text{Var}(\hat{\theta})$ small?

2. Large sample: As $n \to \infty$, does $\hat{\theta}$ "get close" to θ in some way? (And how quickly?)

   - Finite sample distributions are sometimes too complex for direct analysis

To evaluate finite sample properties, we consider the bias, the variance, and the Mean Squared Error (MSE) as measures of the performance of a given estimator.

Definition 15. *The **bias** of an estimator is defined as the difference between the expectation of the estimator and the estimand of interest, i.e.,*

$$bias(\hat{\theta}) = E(\hat{\theta}) - \theta.$$

*If $E(\hat{\theta}) = \theta$ for all $\theta \in \Theta$, then $\hat{\theta}$ is called **unbiased**.*

Definition 16. *The **Variance** of an estimator (a measure of precision) is*

$$Var(\hat{\theta}) = E([\hat{\theta} - E(\hat{\theta})]^2) = E(\hat{\theta}^2) - E(\hat{\theta})^2.$$

Definition 17. *The **Mean squared error (MSE)** of an estimator measures the overall quality of an estimator, and is defined as the squared expectation of the difference between $\hat{\theta}$ and θ:*

$$MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2.$$

The following theorem connects the MSE of a estimator with the mean and the variance of that estimator.

Theorem 18.

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + (bias(\hat{\theta}))^2$$

Note that MSE is a measure of accuracy balancing precision and bias. One of the most commonly considered cases is when bias $= 0$ and variance $=$ minimum. MSE can also be regarded as the risk function $R_{\hat{\theta}}(\theta)$ under squared error loss.

5.2 Consistency

Denote $\hat{\theta}_n = \hat{\theta}(X_1, \ldots, X_n)$ to remind us of dependence on n, and let n get large. We usually want to know

1. $\mathrm{E}[\hat{\theta}_n] \to \theta$ as $n \to \infty$?
2. $\mathrm{Var}[\hat{\theta}_n] \to 0$ as $n \to \infty$?
3. 1. and 2.?
4. $\mathcal{L}(\hat{\theta}_n) \to$ degenerate distribution, with mass 1 at θ?
5. $\hat{\theta}_n$ increasingly concentrates around θ as $n \to \infty$; i.e., $\hat{\theta}_n \xrightarrow{P} \theta$, or $P(\theta - \epsilon < \hat{\theta}_n < \theta + \epsilon)$ as $n \to \infty$ for any $\epsilon > 0$?
6. All of the above, and more?

Suppose $Z_1, Z_2, \ldots$ is sequence of random variables defined on a common probability space with $Z_j \sim F_j$, and suppose Z is defined on the same space.

Convergence in Probability: $Z_n \xrightarrow{P} Z$ if for all $\epsilon > 0$,

$$P(|Z_n - Z| > \epsilon) \to 0 \text{ as } n \to \infty.$$

Equivalently, $Z_n \xrightarrow{P} Z$ if for all $\epsilon > 0$,

$$P(|Z_n - Z| < \epsilon) \to 1 \text{ as } n \to \infty.$$

Chebyshev's Inequality:

$$P(|Y - c| \geq a) \leq \frac{\mathrm{E}[(Y - c)^2]}{a^2}.$$

Thus, a sufficient condition for $Y_n \xrightarrow{P} c$ is that $\mathrm{E}[(Y - c)^2] \to 0$.

Weak Law of Large Numbers (WLLN I): $Z_1, Z_2, \ldots$ are iid, and $\mu = E(Z_i)$ is finite. Then $\bar{Z}_n = n^{-1}(Z_1 + \cdots + Z_n) \xrightarrow{P} \mu$.

Weak Law of Large Numbers (WLLN II): Suppose $X_1, X_2, \ldots$ are random variables with $E(X_i) = \mu_i$, $\mathrm{Var}(X_i) = \sigma_i^2 < \infty$, $\mathrm{Cov}(X_i, X_j) = 0$ for all $i \neq j$. If $\lim_{n\to\infty} \sum \sigma_i^2/n^2 = 0$ then $\bar{X}_n - \bar{\mu}_n \xrightarrow{P} 0$.

Definition 18. *If $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$ for each $\theta \in \Theta$, then $\hat{\theta}_n$ is* **consistent** *for θ.*

Now consider 1-2 from the beginning of Section 3.2. Suppose bias$(\hat{\theta}_n) \to 0$ and Var$(\hat{\theta}_n) \to 0$. Then

$$\mathrm{E}[(\hat{\theta}_n - \theta)^2] = \mathrm{Var}(\hat{\theta}_n) + [E(\hat{\theta}_n) - \theta]^2 \to 0.$$

Thus, $MSE = \mathrm{E}[(\hat{\theta}_n - \theta)^2] \to 0$ as $n \to \infty$.

Then from Chebyshev's inequality, for any fixed $\epsilon > 0$,

$$P(\left|\hat{\theta}_n - \theta\right| \geq \epsilon) \leq \frac{\mathrm{E}[(\hat{\theta}_n - \theta)^2]}{\epsilon^2} \to 0.$$

Hence $\hat{\theta}_n \xrightarrow{P} \theta$.

Note that if $\hat{\theta}_n \xrightarrow{P} \theta$, then it follows from

$$1 = \lim_n P(\left|\hat{\theta}_n - \theta\right| < \epsilon) = P(\theta - \epsilon < \hat{\theta}_n < \theta + \epsilon)$$

for all $\epsilon > 0$ that the limiting distribution of $\hat{\theta}_n$ puts all of its mass on θ (4-5 from the beginning of Section 3.2).

Example: $X_1, \ldots, X_n$ iid, $\mathrm{E}(X_i) = \theta$, $\mathrm{Var}(X_i) = 1$. Consider $\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n$.

Example: Again $X_1, X_2, \ldots$ iid, $\mathrm{E}(X_i) = \theta$, $\mathrm{Var}(X_i) = 1$. Also suppose $Z_n \sim$ Bernoulli$(1/n)$ and independent of the X_i, and that $\hat{\theta}_n = \bar{X}_n + nZ_n$.

Conclusion 1:

$$\mathrm{E}(\hat{\theta}_n) \to \theta \text{ and } \mathrm{Var}(\hat{\theta}_n) \to 0$$
$$\Rightarrow \hat{\theta}_n \xrightarrow{P} \theta,$$

but not the converse.

Conclusion 2: Consistency is a weak (or minimal) property for estimators to satisfy. It does not guarantee that the estimator is very good.

Lemma 1. *If we assume that $Y_1, \ldots, Y_n$ are uniformly bounded, i.e.,*

$$P(|Y_n - c| < M) = 1 \textit{ for all } n,$$

Then,

$$Y_n \xrightarrow{P} c \quad \Rightarrow \quad E(Y_n) \to c, \textit{Var}(Y_n) \to 0.$$

The boundedness condition can be relaxed. A sequence of random variables $Y_1, Y_2, \ldots$ is **uniformly integrable** if

$$\lim_{\alpha \to \infty} \sup_n \mathrm{E}(|Y_n| \, I(|Y_n| > \alpha)) \to 0.$$

Theorem 19. *If $Y_n \xrightarrow{P} Y$ and that Y_n are uniformly integrable, then $E(Y_n) \to E(Y)$.*

(Proof: Some variation on this result is derived in most more advanced probability books, such as Billingsley, 3rd ed, 1995, Thm 25.12)

If Y_n is uniformly integrable, then there is some α^* such that $\sum_n \mathrm{E}(|Y_n| \, I(|Y_n| > \alpha)) < 1$ for $\alpha > \alpha^*$ (since the limit is 0). Thus there are values $\infty > \alpha > \alpha^*$ such that

$$\mathrm{E}(|Y_n|) \leq \alpha P(|Y_n| \leq \alpha) + \mathrm{E}(|Y_n| \, I(|Y_n| > \alpha)) \leq \alpha + 1$$

for all n, so $\sup_n \mathrm{E}(|Y_n|) < \infty$.

Since $\mathrm{E}(|Y_n| \, I(|Y_n| > \alpha)) \leq \frac{1}{\alpha^\delta} \mathrm{E}(|Y_n|^{1+\delta})$, if $\sup_n \mathrm{E}(|Y_n|^{1+\delta}) < \infty$ for some $\delta > 0$, then the Y_n are uniformly integrable.

Theorem 20. *If A_n and B_n are two sequences of random variables satisfying $A_n \xrightarrow{P} a$ and $A_n \xrightarrow{P} b$,*

1. $A_n + B_n \xrightarrow{P} a + b$
2. $A_n - B_n \xrightarrow{P} a - b$
3. $A_n B_n \xrightarrow{P} ab$
4. $A_n / B_n \xrightarrow{P} a/b$ *as long as* $b \neq 0$

Theorem 21. *(Continuous Mapping Theorem) If* $Y_n \xrightarrow{P} c$ *and* g *is continuous at* c *then* $g(Y_n) \xrightarrow{P} g(c)$.

Example: $X_1, \ldots, X_n$ iid Exp(λ).

$$f(x_i; \lambda) = \lambda \exp\{-\lambda x_i\} I(x_i \geq 0).$$

Recall, $\mathrm{E}(X_i) = 1/\lambda$.

For vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^k$, the Euclidean distance between $\boldsymbol{a}$ and $\boldsymbol{b}$ is

$$d_2(\boldsymbol{a}, \boldsymbol{b}) = \left(\sum_{j=1}^{k} (a_j - b_j)^2 \right)^{1/2} = \|a - b\| .$$

Definition 19. *Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots$ be a sequence of random vectors in $\mathbb{R}^k$; $\boldsymbol{X}_n = (X_{1n}, \ldots, X_{kn})$. Also let c be a vector of constants. Then $\boldsymbol{X}_n \xrightarrow{P} c$ if for every $\epsilon > 0$,*

$$\lim_{n \to \infty} P(d_2(\boldsymbol{X}_n, c) > \infty) = 0.$$

Also, $\boldsymbol{X}_n \xrightarrow{P} \boldsymbol{X}$ if $\boldsymbol{X}_n - \boldsymbol{X} \xrightarrow{P} 0$.

Theorem 22. *$X_n \xrightarrow{P} c$ in $\mathbb{R}^k$ if and only if $X_{jn} \xrightarrow{P} c_j$ in $\mathbb{R}$ for $j = 1, \ldots, k$.*

The continuous mapping theorem also holds for random vectors: if $g : \mathbb{R}^k \to \mathbb{R}$ is continuous at c, and $X_n \xrightarrow{P} c$, then $g(X_n) \xrightarrow{P} g(c)$. The proof is essentially the same as in the univariate case.

Note that the results in Theorem 20 can be derived from this, since addition and multiplication are continuous functions from $\mathbb{R}^2 \to \mathbb{R}$.

Consistency of Sample Moments: $X_1, \ldots, X_n$ iid mean μ, $E(|X_i|^k) < \infty$, $\mu_k = \mathrm{E}[(X_i - \mu)^k]$ is the kth central moment. Then $m_k = \sum_{i=1}^n (X_i - \bar{X})^k / n$ is a consistent estimator of μ_k as $n \to \infty$.

In the method of moments, given a parameterization for the moments $\mu_j(\theta)$, $\hat{\theta}$ is determined so that

$$m_1 = \mu_1(\hat{\theta}),\ \ldots,\ m_k = \mu_k(\hat{\theta}).$$

- $m_j \xrightarrow{P} \mu_j(\theta)$
- If each θ_j is a continuous function of $\mu_1, \ldots, \mu_k$, then consistency of $\hat{\theta}_n$ follows from continuous mapping theorem, i.e., $\theta_j = g_j(\mu_1, \ldots, \mu_k)$, at least implicitly

In the one parameter case, using Taylor's theorem,

$$m_1 = \mu_1(\hat{\theta}) = \mu_1(\theta) + \frac{d}{d\theta}\mu_1(\tilde{\theta})(\hat{\theta} - \theta),$$

where $\tilde{\theta}$ is some value between θ and $\hat{\theta}$. Thus if $|d\mu_1(\theta)/d\theta| > \delta > 0$, then

$$\left|\hat{\theta} - \theta\right| = |m_1 - \mu_1(\theta)| \,/\, \left|d\mu_1(\tilde{\theta})/d\theta\right| \leq |m_1 - \mu_1(\theta)| \,/\delta \xrightarrow{P} 0.$$

5.3 Mean Squared Error

Back to Mean Squared Error (MSE):

$$MSE(\hat{\theta}) = \mathrm{E}[\hat{\theta} - \theta]^2.$$

<u>Example</u>: $X_1, \ldots, X_n$ iid $\mathrm{E}(X_i) = \mu$, $\mathrm{Var}(X_i) = \sigma^2$. $\bar{X} = (X_1 + \cdots + X_n)/n$. Let S^2 be the usual sample variance estimate of σ^2, i.e.,

$$S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2,$$

and

$$\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2 = \frac{n-1}{n}S^2.$$

Which one is better in terms of MSE in general?

1. Since S^2 is location invariant, without loss of generality assume $\mu = 0$ (Or replace X_i with $Y_i = X_i - \mu$ throughout)

2. Assume distribution of X_i is symmetric, so $\mathrm{E}(X_i^k) = 0$ for k odd

$$\begin{aligned}\mathrm{Var}(S^2) &= \frac{1}{(n-1)^2}\left\{\mathrm{Var}(\sum_i X_i^2) + n^2\mathrm{Var}(\bar{X}^2) - 2\,\mathrm{Cov}(\sum_i X_i^2, n\bar{X}^2)\right\} \\ &= \frac{1}{(n-1)^2}\left\{n\mathrm{Var}(X_i^2) + n^2\mathrm{Var}(\bar{X}^2) - 2\mathrm{Var}(X_i^2)\right\},\end{aligned}$$

since $\bar{X}^2 = n^{-2}\left(\sum X_i^2 + \sum_{i\neq j} X_i X_j\right)$, and using independence and the fact that $\mathrm{E}(X_i^k) = 0$ for k odd, from assumptions 1 and 2.

- Possible to express $\mathrm{Var}(\bar{X}^2)$ in terms of $\mathrm{Var}(X_i^2)$ and $\sigma^2 = \mathrm{Var}(X_i)$, to give a formula for a general symmetric distribution in terms of $\mathrm{Var}(X_i^2)$ and σ^2

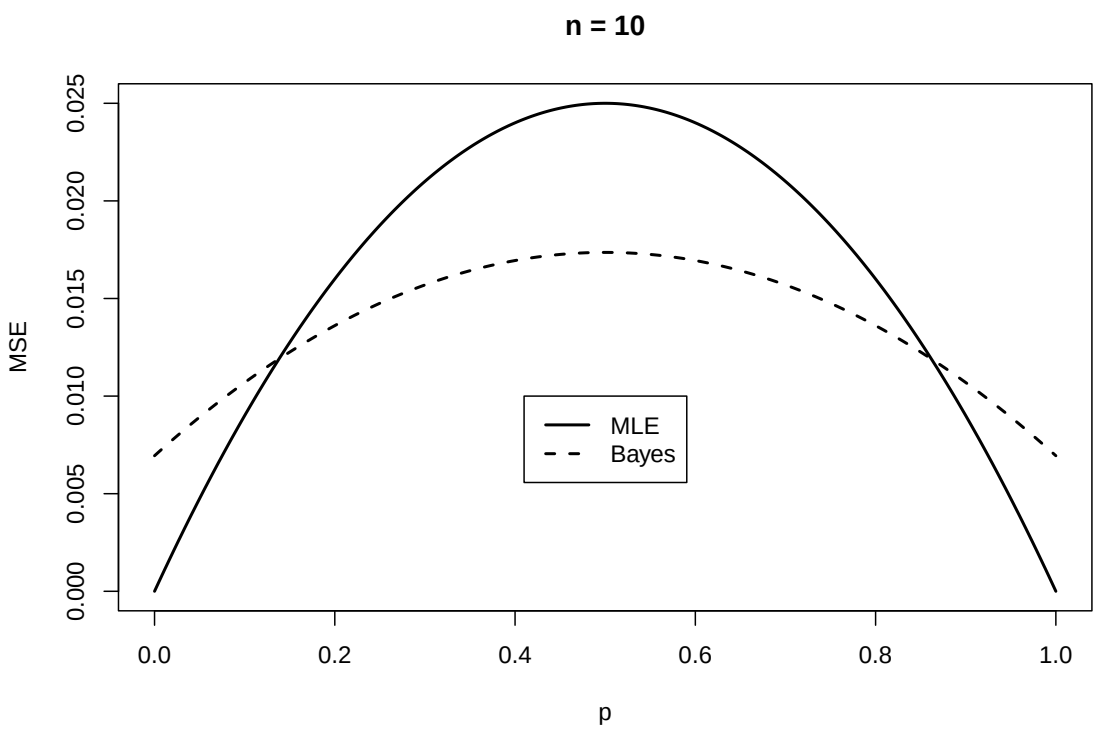
- If in addition, $X_i \sim N(0, \sigma^2)$, then $\mathrm{Var}(X_i^2) = 2\sigma^4$ and $\mathrm{Var}(\bar{X}^2) = 2\sigma^4/n^2$ (since $\bar{X} \sim N(0, \sigma^2/n)$), so in this case:

Ideally, we want to seek T^* such that $MSE(T^*;\theta) \leq MSE(T^*;\theta)$ for all θ and T. However,

- Dominating all possible estimators not possible
- Need to restrict to a class of estimators that meet some minimal requirements across the parameter space
- E.g., unbiased estimators

Example of MSE: Let $X_1, \cdots, X_n$ are iid Bernoulli(p). Consider two estimators: $\bar{X}$ vs. $(\sum_{i=1}^n X_i + 1)/(n+2)$.

Figure 2: MSE of MLE vs. Bayes



Definition 20. *Consider 2 estimators, say $\hat{\theta}$ and $\tilde{\theta}$, of θ. If*

$$MSE(\hat{\theta}) \leq MSE(\tilde{\theta}) \text{ for all } \theta \in \Theta$$

and

$$MSE(\hat{\theta}) < MSE(\tilde{\theta}) \text{ for at least one } \theta \in \Theta,$$

then $\tilde{\theta}$ is said to be **inadmissible** *(wrt squared error loss). If $\tilde{\theta}$ is not inadmissible it is called* **admissible**.

That is, with respect to MSE, $\tilde{\theta}$ is inadmissible if there exists $\hat{\theta}$ such that $\tilde{\theta}$ sometimes worse and never better than $\hat{\theta}$; $\tilde{\theta}$ is admissible if there is no other estimator that always better than $\tilde{\theta}$. From previous example, for iid $X_i \sim N(\mu, \sigma^2)$, S^2 is inadmissible because there is another estimator ($\hat{\sigma}^2$) with smaller MSE.

Some times, we use inadmissible estimators because

- the difference in MSE is small
- they have other attractive properties (e.g. unbiased, standard distributions)

5.3.1 The Mean and the Median

In this section, we compare the mean and median estimators under different data generation processes. Suppose $X_1, \cdots, X_n$ are iid, and the pdf of X_i is given by $f(x - \mu)$, where f is a symmetric density ($f(\mu + x) = f(\mu - x)$ for all x), and $-\infty < \mu < \infty$, Consider 4 cases:

1. Normal: $f(x) = (2\pi)^{-1/2} \exp(-x^2/2)$,
2. Logistic: $f(x) = \exp(x)/[1 + exp(x)]^2$,
3. Double Exponential: $f(x) = \exp(-|x|)/2$,
4. Cauchy: $f(x) = [\pi(1 + x^2)]^{-1}$.

In the first 3 cases, $\mathrm{E}(\bar{X}) = \mu$, $\mathrm{Var}(\bar{X}) = \mathrm{Var}(X_i)/n$, where $\mathrm{Var}(X_i) = 1$ for the normal, $\mathrm{Var}(X_i) = \pi^2/3$ for the logistic, and $\mathrm{Var}(X_i) = 2$ for the double exponential. For the Cauchy distribution, $\mathrm{E}(\bar{X})$ does not exist because $\mathrm{E}(X_i) = \infty$.

For the median (M), assume n is odd, so $M = X_{(n+1)/2}$. Then the density of median is

$$q(m;\mu) = n\binom{n-1}{(n-1)/2} f(m-\mu)[F(m-\mu)(1-F(m-\mu))]^{(n-1)/2},$$

where F is the cdf corresponding to the pdf f. q is symmetric about μ since $F(-\mu) = 1 - F(\mu)$. Then

$$\begin{aligned}\mathrm{E}(M) &= \int_{-\infty}^{\infty} n\binom{n-1}{(n-1)/2} mf(m-\mu)[F(m-\mu)(1-F(m-\mu))]^{(n-1)/2}dm \\ &=\end{aligned}$$

Thus in all 4 cases the sample median m is an unbiased estimator of μ.

For the variance of the sample median, we have

$$\begin{aligned}\text{Var}(M) &= \int_{-\infty}^{\infty} n \binom{n-1}{(n-1)/2} (m-\mu)^2 f(m-\mu)[F(m-\mu)(1-F(m-\mu))]^{(n-1)/2} dm \\ &=\end{aligned}$$

The above integration can be evaluated numerically.

From the above example, we should notice that

1. True distribution usually unknown
2. Seldom a clear choice of which estimator to use
3. Other estimators could be considered (e.g. MLE from logistic distribution)

6 Evaluating Estimators II: Best Unbiased Estimation

Outline:

- Cramer-Rao Inequality
- Estimators of 0
- Rao-Blackwell
- Complete Families

6.1 Cramer-Rao Inequality

In this section, we will focus on *unbiased estimators*.

Definition 21. *If $\hat{\theta}$ and $\tilde{\theta}$ are any unbiased estimators of θ whose variances exist, then*

$$RE(\hat{\theta};\tilde{\theta}) \equiv \frac{Var(\tilde{\theta})}{Var(\hat{\theta})}$$

is called the **relative efficiency** *of $\hat{\theta}$ to $\tilde{\theta}$.*

RE provides a useful way to compare 2 unbiased estimator, i.e. $RE(\hat{\theta};\tilde{\theta}) > 1$ means $\hat{\theta}$ has smaller variance (indeed smaller MSE). However, it doesn't tell us if the better one is optimal in some way. Thus, our next goal is to find bounds on variance of unbiased estimators.

Let's first study two important results.

Theorem 23. *Suppose* $X \sim f(x;\theta), \theta \in \Theta$. *Assume*

$$\frac{\partial}{\partial\theta}\int f(x;\theta)dx = \int \frac{\partial}{\partial\theta} f(x;\theta)dx. \tag{6}$$

Then

$$E_\theta\left[\frac{\partial}{\partial\theta}\log f(X;\theta)\right] = 0 \text{ for all } \theta.$$

Note1: $f(x;\theta)$ is the pdf of X but $f(X;\theta)$ is a random variable.
Note2: Intuitively, a sufficient condition for (6) to be satisfied is that the partial derivative $\frac{\partial}{\partial\theta}f(X;\theta)$ needs to be bounded by a integrateable function.

Example: log-likelihood. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim f(x_i; \theta)$.

Theorem 24. *Suppose* $X \sim f(x;\theta), \theta \in \Theta$. *Assume*

$$\frac{\partial^2}{\partial\theta^2}\int f(x;\theta)dx = \int \frac{\partial^2}{\partial\theta^2} f(x;\theta)dx. \tag{7}$$

Then

$$E_\theta\left[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\right] = -E\left[\left(\frac{\partial}{\partial\theta}\log f(X;\theta)\right)^2\right]$$

Note: (7) holds for exponential families.

Definition 22. *An estimator W of a parameter, say $\tau(\theta)$, is called the best unbiased estimator, or* **uniform minimum variance unbiased estimator (UMVUE)** *of $\tau(\theta)$ if*

$$E_\theta(W) = \tau(\theta) \text{ for all } \theta \in \Theta$$

and

$$Var_\theta(W) \leq Var_\theta(Q)$$

for any other unbiased estimator Q of $\tau(\theta)$.

Here, "uniform" means for all $\theta \in \Theta$. Then, we need to consider the following questions:

- How do we find (if exist) a UMVUE?
- How can we tell if a candidate estimator is a UMVUE?

Theorem 25. Cramer-Rao Inequality. *Suppose $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim f(x_i;\theta)$, $\theta \in \Theta$, and also suppose (6) in Theorem 23 is satisfied. If $W = W(\boldsymbol{X})$ be some estimator such that*

$$\frac{\partial}{\partial\theta} E_\theta(W) = \int \frac{\partial}{\partial\theta}[W(\boldsymbol{x}) f(\boldsymbol{x};\theta)] d\boldsymbol{x},$$

then

$$Var(W) \geq \frac{\left[\frac{\partial}{\partial\theta} E_\theta(W)\right]^2}{E_\theta\left(\left[\frac{\partial}{\partial\theta}\log f(\boldsymbol{X};\theta)\right]^2\right)}.$$

In addition, if (7) in Theorem 24 is also satisfied, then

$$Var(W) \geq \frac{\left[\frac{\partial}{\partial\theta} E_\theta(W)\right]^2}{nE_\theta\left[-\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta)\right]}.$$

To proof this Cramer-Rao inequality, we need the following so-called *Cauchy-Schwarz inequality*:
If $\mathrm{E}(X_1^2) < \infty$ and $\mathrm{E}(X_2^2) < \infty$, then

$$|\mathrm{E}(X_1 X_2)| \leq \sqrt{\mathrm{E}(X_1^2)\mathrm{E}(X_2^2)},$$

with equality if and only if $X_1 = aX_2$ for some constant a.

Example: Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim \text{Poisson}(\lambda)$.

Notes:

- Cramer-Rao inequality (Theorem 25) avoids any mention of what W is estimating. It just gives a lower bound for the variance of any unbiased estimator.
- The bound is not always attained: maybe no statistic T that is unbiased for $\mathrm{E}(W)$ such that

$$\mathrm{Var}(T) = \frac{\left[\frac{\partial}{\partial\theta}\mathrm{E}_\theta(W)\right]^2}{\mathrm{E}_\theta\left(\left[\frac{\partial}{\partial\theta}\log f(\boldsymbol{X};\theta)\right]^2\right)}.$$

- Unbiased estimators may not exist for a given function of parameter.

Example: no unbiased estimator exists. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim \mathrm{Binomial}(n, p)$, and we want to estimate $\tau(p) = 1/p$.

Remark 1: If $\hat{\theta}$ doesn't attain the Cramer-Rao lower bound, it doesn't follow that it can't be an UMVUE.

Remark 2: Cramer-Rao lower bound holds for any functions of θ.

Remark 3: If $\frac{\partial}{\partial\theta}\mathrm{E}_\theta(W) = \int \frac{\partial}{\partial\theta}[W(\boldsymbol{x})f(\boldsymbol{x};\theta)]d\boldsymbol{x}$ doesn't hold, the Cramer-Rao lower bound may not be a lower bound on the variance.

When $\boldsymbol{\theta}$ is k-dimensional,

$$\frac{\partial}{\partial\boldsymbol{\theta}}\log f(\boldsymbol{X};\boldsymbol{\theta}) = \left(\frac{\partial}{\partial\theta_1}\log f(\boldsymbol{X};\boldsymbol{\theta}), \cdots, \frac{\partial}{\partial\theta_k}\log f(\boldsymbol{X};\boldsymbol{\theta})\right)^T.$$

Generally, $\mathrm{E}\left(\frac{\partial}{\partial\boldsymbol{\theta}}\log f(\boldsymbol{X};\boldsymbol{\theta})\right) = 0$, which basically follows from applying the Theorem 23 to each component. The **information matrix** $I(\boldsymbol{\theta})$ is the variance-covariance matrix of $\frac{\partial}{\partial\boldsymbol{\theta}}\log f(\boldsymbol{X};\boldsymbol{\theta})$, which is the matrix with $(i,j)^{th}$ components $\mathrm{E}\left(\frac{\partial\log f(\boldsymbol{X};\boldsymbol{\theta})}{\partial\theta_i}\frac{\partial\log f(\boldsymbol{X};\boldsymbol{\theta})}{\partial\theta_j}\right)$. It also equals to $\mathrm{E}\left(-\frac{\partial^2\log f(\boldsymbol{X};\boldsymbol{\theta})}{\partial\theta_i\partial\theta_j}\right)$ if (7) holds (e.g. exponential families). The lower bound on the variance of unbiased estimator W of $\tau(\boldsymbol{\theta})$ is

$$\mathrm{Var}_{\boldsymbol{\theta}}(W) \geq \boldsymbol{d}^T[I(\boldsymbol{\theta})]^{-1}\boldsymbol{d},$$

where

$$\boldsymbol{d} = \left(\frac{\partial\tau(\boldsymbol{\theta})}{\partial\theta_1}, \cdots, \frac{\partial\tau(\boldsymbol{\theta})}{\partial\theta_k}\right)^T.$$

Example: Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim N(\mu, \sigma^2)$. Here $\boldsymbol{\theta} = (\mu, \sigma^2)$, and we want to estimate $\tau(\boldsymbol{\theta}) = \sigma^2$.

In the above example, we show that S^2 does not attain the Cramer-Rao lower bound, but is there a better unbiased estimator?

Corollary 2. *(Attainment). Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $f(x_i; \theta)$. Under the conditions for the Cramer-Rao inequality, an unbiased estimator W of $\tau(\theta)$ attains the Cramer-Rao lower bound if and only if $\frac{\partial}{\partial\theta} l(\theta; \boldsymbol{x}) = a(\theta)[W - \tau(\theta)]$ for some $a(\theta)$.*

Example: Exponential distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim$ Exponential(λ).

Example: Normal distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim N(\mu, \sigma^2)$.

Theorem 26. *Let W be a statistic with $Var(W) < \infty$. If W is a UMVUE of $\tau(\theta)$, then W is the* **unique** *UMVUE of $\tau(\theta)$.*

In summary,

- Cramer-Rao inequality is useful for finding bounds on variance of estimates
- Corollary can help identify UMVUE's.
- There may be no estimator that achieves Cramer-Rao lower bound
- If W is UMVUE and $\text{Var}(W) < \infty$, W is unique UMVUE

6.2 Estimators of 0

An estimator $U(\boldsymbol{X})$ is an unbiased estimator of 0 is $\text{E}(U(\boldsymbol{X})) = 0$ for all $\theta \in \Theta$. If an unbiased estimator W of $\tau(\theta)$ is correlated with some unbiased estimator of 0, then this unbiased estimator can be used to improve W.

Specifically, suppose $\text{E}(W) = \tau(\theta)$, $\text{Var}(W) < \infty$, $\text{E}(U(\boldsymbol{X})) = 0$, $\text{Var}(U(\boldsymbol{X})) < \infty$, and $\text{Cov}(W, U) \neq 0$. $W + bU$ is also an unbiased estimator of $\tau(\theta)$. Then

Thus, finite variance UMVUE's must be uncorrelated with finite variance unbiased estimators of 0.

Theorem 27. *If $E(W(\boldsymbol{X})) = \tau(\theta)$ and $Var(W(\boldsymbol{X})) < \infty$, then $W(\boldsymbol{X})$ is best unbiased if and only if $W(\boldsymbol{X})$ is uncorrelated with all finite variance unbiased estimators of 0.*

Example: $X_1, \cdots, X_n$ are iid $N(\mu, 1)$. Let $W = \sum_{i=1}^{n-1} X_i/(n-1)$, $U = X_n - W$.

In summary,

- An unbiased estimator of 0 is random noise; a good estimator should not be improved by adding random noise to it.
- Difficult to verify that an estimator is uncorrelated with all unbiased estimators of 0; easier to verify that an estimator is correlated with one unbiased estimator of 0 (i.e., use this to verify that an estimator is not best unbiased).
- This result says that if there are NO unbiased estimators of 0, then an unbiased estimator is best.
- No functions of a complete and sufficient statistic can be an unbiased estimator of 0.

6.3 Rao-Blackwell

Theorem 28. (Rao-Blackwell). *For a sample $\boldsymbol{X}$, suppose $W(\boldsymbol{X})$ is unbiased for $\tau(\theta)$ and $T(\boldsymbol{X})$ is sufficient for θ. Define $Y = E(W|T)$. Then*

1. *Y is a statistic (not a function of θ).*
2. *$E(Y) = \tau(\theta)$ for all θ (Y is an unbiased estimator for $\tau(\theta)$).*
3. *$Var(Y) \leq Var(W)$ for all θ.*

Example: X_1, X_2 iid Poisson(θ). Consider $W = X_1, T = X_1 + X_2 \sim$ Poisson(2θ).

Notes:

- Result holds if θ and T are vectors. Only requires $\tau(\theta)$ and W be scalars.
- From the above example, in theory, we can improve W by conditioning on T; however, in practice, computing the conditional expectation can be difficult.
- Idea behind Rao-Blackwell is to improve an unbiased estimate by reducing "noise" in it through conditioning on a sufficient statistic.

6.4 Complete Families

In this section, we study a very powerful theorem and corollary (Lehmann-Scheffe). This theorem can help us get UMVUEs through complete and sufficient statistics.

Theorem 29. *Suppose $\boldsymbol{X} \sim f(\boldsymbol{x};\theta)$, $\theta \in \Theta$, and $\tau(\theta)$ be some scalar function of θ. Let $T = T(\boldsymbol{X})$ be a complete and sufficient statistic and $W = W(\boldsymbol{X})$ be an unbiased estimator of $\tau(\theta)$. Then*

$$Y = E(W|T) \text{ is UMVUE of } \tau(\theta).$$

Also, if $Var(Y) < \infty$, Y is the unique UMVUE.

Corollary 3. *If T is a complete and sufficient statistic, then for any scalar function $g(\cdot)$, $g(T)$ is the UMVUE of $E[g(T)]$.*

These theorem and corollary suggest two ways of find a UMVUE of $\tau(\theta)$ when a complete and sufficient statistic T exists:

1. Find any unbiased estimator, say W, of $\tau(\theta)$, and use Lehmann-Scheffe Theorem to conclude that $\mathrm{E}(W|T)$ is UMVUE.

2. Find a function $g(\cdot)$ such that $\mathrm{E}[g(T)] = \tau(\theta)$ and use the corollary to conclude $g(T)$ is UMVUE.

Example: Poisson distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim$ Poisson(θ).

Example: Binomial distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim$ Binomial(k, p), where k is known and $p \in (0, 1)$. Consider estimating

$$\tau(\theta) = P(X_i = 1) =$$

Example: Exponential distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim$ Exponential(θ).

Example: Bernoulli distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim$ Bernoulli(p). Consider $\tau(p) = p^2$.

Example: Normal distribution. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, X_i iid with $X_i \sim N(\mu, \sigma^2)$. 1. Find the UMVUE of μ; 2. Find the UMVUE of σ^2.

Let $X_1, \cdots, X_n$ iid, $X_i \sim$ exponential family with pdf (pmf)

$$h(x)c(\theta)\exp\left\{\sum_{j=1}^{k} w_j(\theta)t_j(x)\right\}.$$

By Theorem 15, if $\mathcal{H} \equiv \{(w_1(\theta), \cdots, w_k(\theta))|\theta \in \Theta\}$ contains an open rectangle in $\mathbb{R}^k$, then $T = (T_1, \cdots, T_k)$ is a complete and sufficient statistic, where $T_j = \sum_{i=1}^{n} t_j(x_i)$.

Thus, for any exponential family where $\mathcal{H}$ contains an open rectangle,

1. Any function of T that has an expectation is the UMVUE of its expectation (unique if its variance is finite).
2. Any function $\tau(\theta)$ that has an unbiased estimator W has a UMVUE, which can be obtained through $\mathrm{E}(W|T)$.

Finally, a UMVUE does not always exist because

1. There may not even be unbiased estimators.
2. There may be unbiased estimates, but not one that is uniformly best.